

Subject-Normalized Stress-State Discovery from Wearable Biosignals

02582 Computational Data Analysis | Case 2 | s252646

Research question: After removing subject baselines, do HR/TEMP/EDA features contain generalizable stress-phase structure, and how strongly do they relate to affect ratings?

1. Data and Preprocessing

EmoPairCompete
26 participants

Preprocess
median impute

Subject z-score
baseline-aware

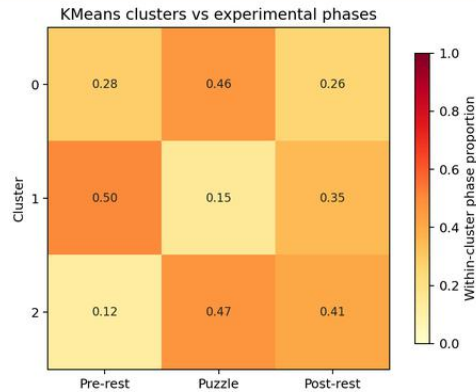
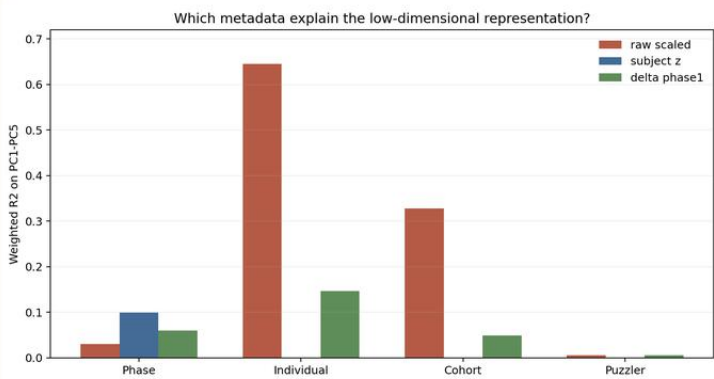
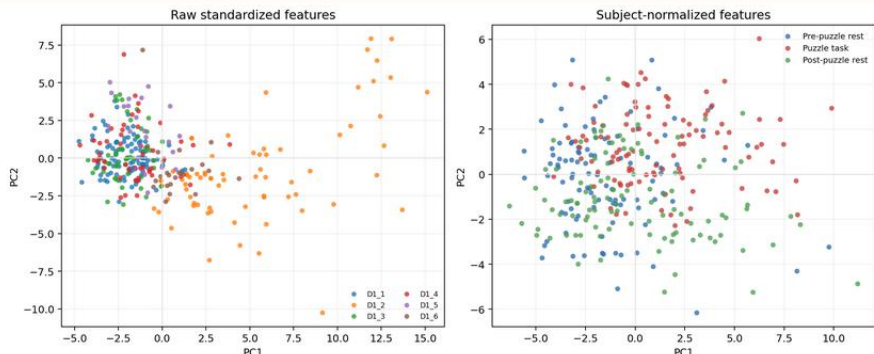
PCA + KMeans
latent states

CCA
cross-modal

UMAP + t-SNE
non-linear check

- Data: 312 samples = 26 individuals x 4 rounds x 3 phases.
- Inputs: 51 HR/TEMP/EDA features; 11 self-report variables used for CCA/evaluation.
- Validation: GroupKFold split by individual; no sample-level leakage.
- Goal: discover representations, not maximize a supervised classifier.

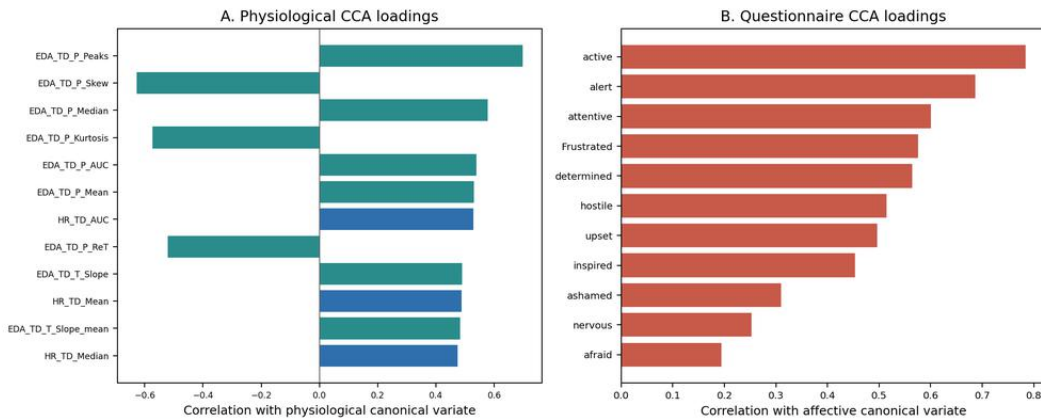
2. PCA: Baselines Dominate



- Raw PC1-PC5 variance is mostly individual identity ($R^2=0.645$) and cohort ($R^2=0.328$).
- Experimental phase explains only $R^2=0.030$ in raw PCA, increasing to $R^2=0.099$ after subject normalization.
- KMeans does not recover clean latent phases: best ARI=0.075 after subject normalization.

3. CCA: Physiology-Affect Coupling

CCA: physiology-affect coupling, in-sample $r=0.477$, GroupKFold $r=0.362 \pm 0.128$



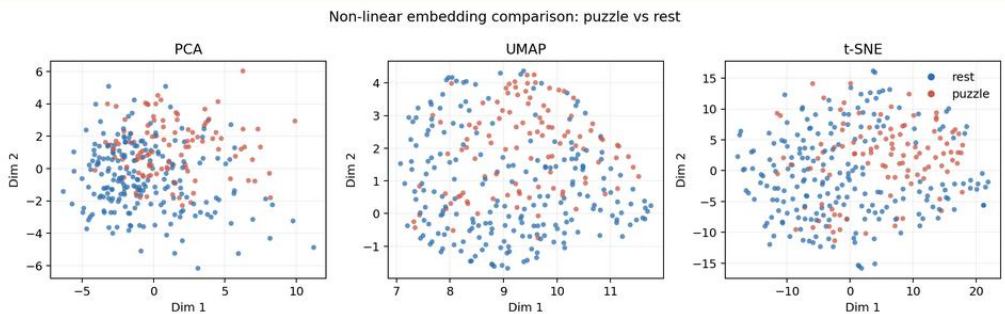
0.477
In-sample canonical correlation

0.362
GroupKFold r across held-out subjects

$p=0.005$
Subject-block permutation test

- CCA links physiological variation to self-reported affect, but the held-out-subject estimate is moderate.
- Top physiological contributors are mainly EDA phasic features, especially peaks, median, AUC, and skewness.
- The affective axis loads on active, alert, attentive, frustrated, and determined: more arousal-like than pure valence.

4. PCA vs UMAP vs t-SNE: Non-linear Comparison



Method	Silhouette	kNN acc.
PCA	0.107	0.737
UMAP	0.069	0.769
t-SNE	0.058	0.763

Non-linear embeddings do not clearly improve phase geometry. UMAP gives slightly higher kNN accuracy, but lower silhouette; t-SNE is visually mixed. PCA remains the most interpretable representation.

Key Take-aways

1. Individual/cohort baselines dominate raw wearable features.

2. EDA-phasic activity is the strongest cross-modal signal carrier.

3. CCA finds moderate arousal-like physiology-affect coupling.

4. No robust unsupervised separation of emotional states.